



Summer of Science (SoS) 2026 – End Term Presentation

GeoQuery: An AI-Powered Natural Language GIS Query System

Large Language Models

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Motivation

- A **Geographic Information System (GIS)** is a computer-based system designed for capturing, storing, managing, analyzing, and visualizing geographically referenced data.
- Traditional GIS requires
 - Knowledge of SQL
 - Knowledge of GIS software
 - Complex spatial queries
 - Technical expertise
- Can we query GIS data like ChatGPT?
 - For Example, "**Show hospitals in Hyderabad**", instead of writing SQL.



Objectives

- Develop an AI-powered GIS query system
- Build a spatial database for India
- Use LLMs to understand user queries
- Generate spatial SQL automatically
- Display results on an interactive map
- Export retrieved data

Study Area



Datasets

Source

- OpenStreetMap
- Geofabrik India

Administrative Boundaries

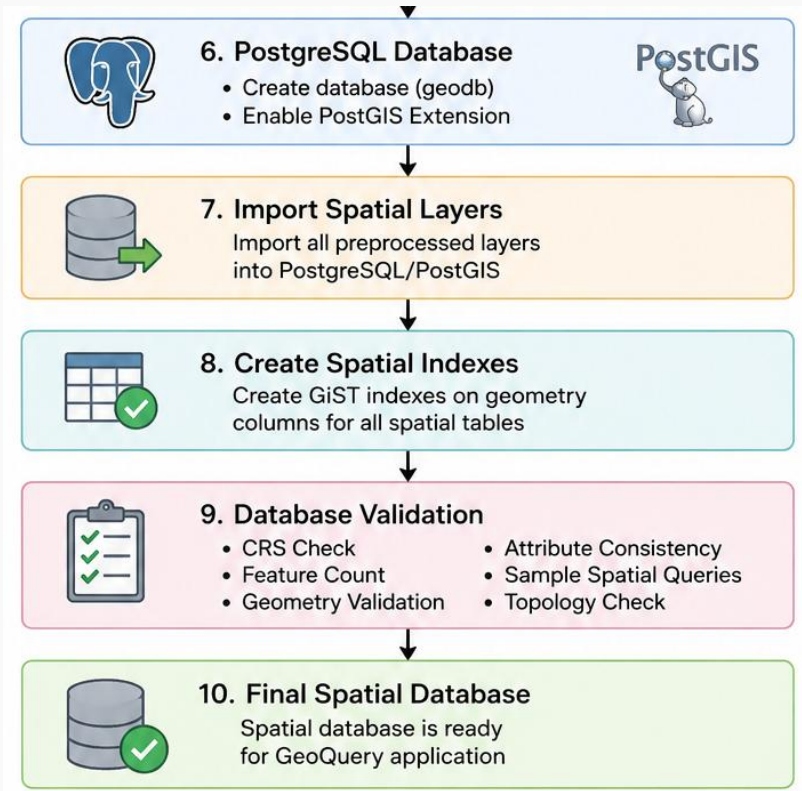
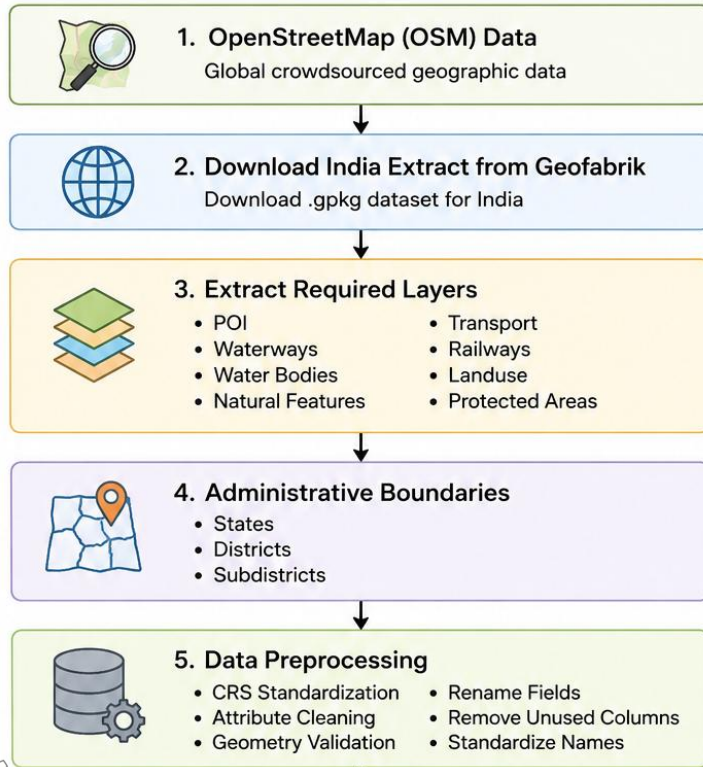
- States
- Districts
- Subdistricts

Layers Imported

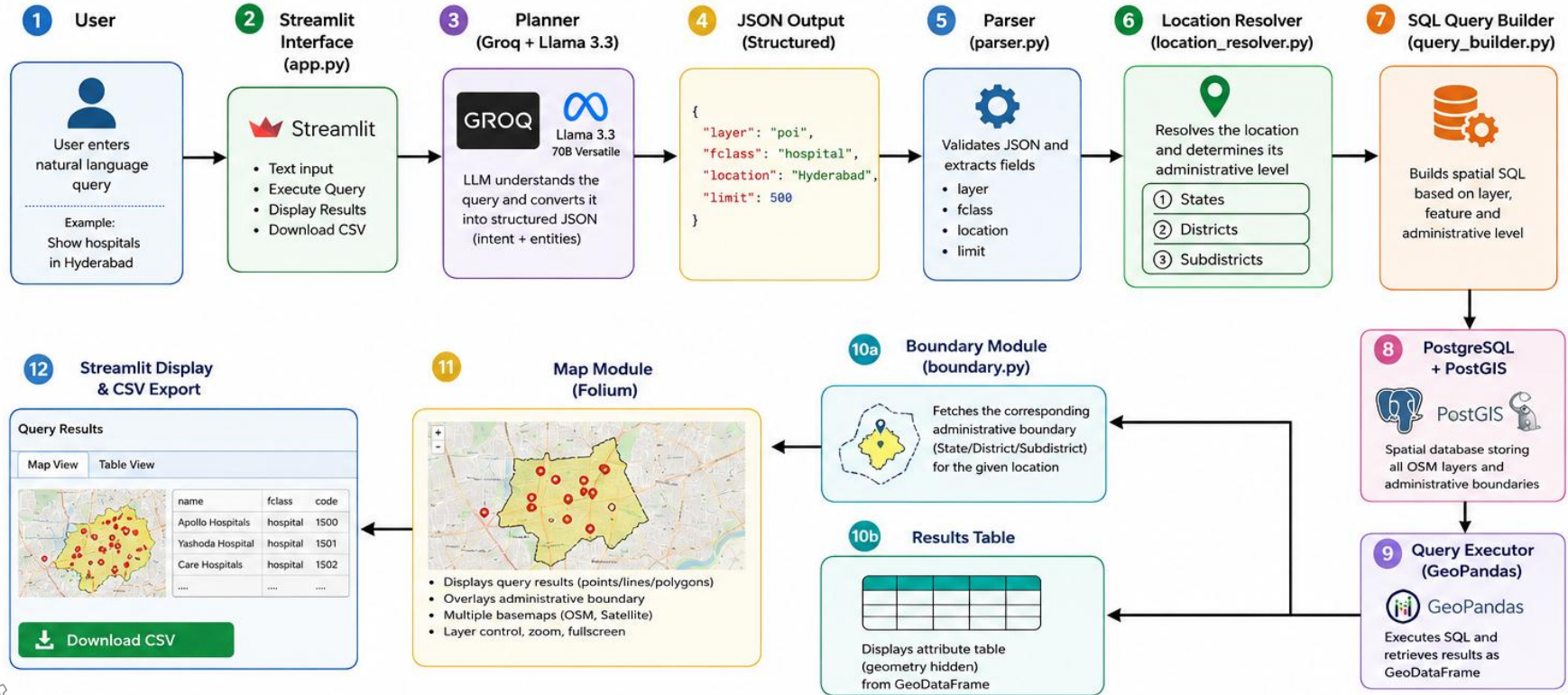
- POI
- Waterways
- Water
- Railways
- Landuse
- Natural
- Transport
- Protected Areas



Database Preparation



System Architecture



Query Processing

User: *"Show hospitals in Hyderabad"*



Planner:

```
{  
  "layer": "poi",  
  "fclass": "hospital",  
  "location": "Hyderabad"  
}
```



SQL Builder



Results



PostGIS Query



Application's UI





Ask a GIS Question

Example: Show hospitals in Hyderabad

Execute Query

Clear

Results

Results will appear here.

Data source: Geofabrik OpenStreetMap • Data as of 10 July 2026

Deploy



+

-



☐ OpenStreetMap

☒ Google Satellite

300 km

200 mi

Leaflet | © OpenStreetMap contributors, Google

Information

Information will appear here.

Demonstration

Example Queries

- Show hospitals in Hyderabad
- Show rivers in Telangana
- Show railways in Goa
- Show restaurants in Delhi
- Show schools in Mumbai



Results

- ✓ Natural language understanding
- ✓ Automatic state/district/subdistrict detection
- ✓ Interactive visualization
- ✓ CSV export
- ✓ Administrative boundary display
- ✓ Multiple GIS layers



Limitations & Future Work

Limitations

- English only
- Depends on OSM
- Internet required
- Vector data only

Future Work

- Raster support
- Google Earth Engine
- Voice queries
- Routing
- Buffer analysis
- Spatial relationships
- RAG integration



Conclusion

GeoQuery successfully

- combines AI with GIS
- converts natural language into spatial SQL
- retrieves spatial data from PostGIS
- visualizes results interactively
- improves GIS accessibility

Thank You

